

CODY MUMFORD

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 7+ years building full-stack web applications with a front-end focus across healthcare and energy — sectors whose systems are critical infrastructure. Deep experience in TypeScript, component-based architectures (Angular, Blazor, Flutter), reactive state management (RxJS, services, ChangeNotifier), and shared UI component libraries. Proven track record of owning features from concept through delivery, simplifying legacy complexity, and leveraging AI-assisted development tools to accelerate engineering velocity and quality.

SKILLS

Programming	TypeScript, C#, JavaScript, Dart, Java, C++
Front-End	Angular, Blazor, Flutter, HTML, CSS, Piral (micro-frontends)
State Management	RxJS, Angular services, Flutter ChangeNotifier
Component Libraries	Piral shared micro-frontend components, NuGet shared UI libraries, reusable design-system patterns
Testing	Jasmine, Karma, xUnit, Moq
Backend / APIs	ASP.NET, .NET Core, Web API, GraphQL, WCF, REST
Database	SQL Server, PostgreSQL, EF Core, SQLite
DevOps	Git, Azure DevOps, GitHub Actions, Kubernetes, Docker, IIS, GHCR
AI-Assisted Dev	GitHub Copilot, Claude Code, Pi, Ollama
OS	Windows, Linux, macOS

WORK EXPERIENCE

Homecare Homebase: Software Engineer II

Remote / Aug 2022 – Present

TypeScript, C#, .NET, Blazor, Piral, Angular, RxJS, SQL Server, GraphQL, Kubernetes, Docker, Azure DevOps, GitHub Copilot

- Own the design, implementation, and evolution of scheduling, billing, and compliance features from concept through delivery in a platform used by home health and hospice agencies nationwide — systems critical to patient care delivery.
- Contributed to a legacy Windows Forms → modern web migration leveraging GitHub Copilot, achieving 100% feature parity across the web frontend and GraphQL backend while reducing technical complexity.
- Helped onboard a new team to the development processes and tech stack, accelerating their ramp-up and integration into the engineering organization.
- Built and maintained a shared Blazor component library consumed by multiple frontends, establishing design-system consistency across teams and enabling reusable, composable UI components for scheduling, billing, and compliance workflows.
- Delivered Blazor micro-frontend components within the Piral framework, enabling modular feature rollouts across the customer-facing web application.
- Collaborated with Product and Design to translate ambiguous requirements into well-scoped features, iterating from wireframes through delivery with continuous feedback loops.
- Created GitHub Copilot agents and skills to streamline team processes, improving engineering effectiveness and reducing repetitive tasks across multiple frontends.
- Architected and deployed containerized web services (Kubernetes, IIS, WCF) packaged as shared NuGet libraries consumed by multiple in-house applications, reducing duplication and improving maintainability.
- Managed end-to-end build and deployment pipelines for customer databases, services, and back-office applications, improving reliability and developer experience.

TypeScript, C#, .NET Core, Angular, RxJS, ESLint, SQL Server, REST APIs

- Designed and implemented Angular components and services with RxJS-based reactive state management powering user-facing forms backed by .NET Core REST APIs for the oil and gas industry — critical infrastructure data.
- Partnered with product stakeholders to define and ship Web API endpoints converting complex industry abstraction models into downloadable Excel workbooks and PDFs.
- Identified and resolved a performance bottleneck in Excel workbook generation, achieving a 30% improvement using JetBrains dotTrace and dotMemory to profile and refactor data access methods — reducing system complexity.
- Implemented secure user authentication using .NET Core, Microsoft Identity, and IdentityServer4.
- Developed a .NET Core Worker Service for asynchronous document download integration, simplifying the client-side experience.

PROJECTS

Kohera: Cross-Platform Matrix Chat Client

github.com/Quantumheart/Kohera | kohera.quantum-matrix.xyz | Shipped

Flutter, Dart, TypeScript, Matrix SDK, SQLite, Docker, GitHub Actions, WebRTC

- Architected and shipped a full-featured cross-platform chat client with end-to-end encryption, cross-signing, and key backup using Rust-based Vodozemac bindings.
- Engineered adaptive, responsive layouts (mobile, tablet, desktop) with a Discord/Slack-style space rail, rich message rendering, WebRTC calling, and OpenGraph link previews.
- Implemented service-based state management with ChangeNotifier for reactive UI updates across a complex real-time messaging domain.
- Built a reusable widget architecture with consistent design-system patterns across all target platforms.
- Shipped to Web, Linux, macOS, iOS, and Windows with CI/CD via GitHub Actions; containerized web deployment with Docker on GHCR.

Bairn: Privacy-First P2P Baby Monitor

codeberg.org/Quantumheart/bairn | In Progress

Flutter, Dart, WebRTC, Bonjour/mDNS, QR Pairing

- Built a peer-to-peer baby monitor that streams live video directly between devices over WebRTC — no cloud relay, no third-party storage, no account-tied surveillance.
- Implemented two signaling paths: QR code pairing for manual exchange and automatic LAN discovery via Bonjour/mDNS for zero-config same-network connection.
- Engineered a transport-agnostic WebRTC handshake using non-trickle ICE with all candidates embedded in the SDP, enabling any signaling transport (QR, TCP, WebSocket) without changing the peer connection logic.
- Designed a nursery mode with screen wakelock, minimum-brightness overlay, and audio-only background fallback — validated through an iOS background-capture spike confirming 36 minutes of steady 30 fps capture with nominal thermals.